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Inventor(s)	Yoshida; Taiki et al.

Information processing method

Abstract

An information processing method includes receiving first information indicating a travel plan of a user, acquiring, for each on-demand vehicle service, second information indicating boarding/alighting points and operation status of a vehicle, determining and presenting, based on the first and second information, a recommended route for transferring between first and second vehicles belonging to first and second on-demand vehicle services, and making a reservation with the first and second on-demand vehicle services in accordance with the recommended route. In the method, a first boarding/alighting point for boarding the first vehicle, a candidate transfer point for transferring from the first to the second vehicle, and a second boarding/alighting point for alighting from the second vehicle are determined, at least one candidate route from the first to the second boarding/alighting point via the candidate transfer point is determined, and based on the evaluation of each candidate route, the recommended route is determined.

Inventors: Yoshida; Taiki (Tokyo-to, JP), Nose; Kenta (Kobe, JP), Hashimoto; Masaaki (Tokyo-to, JP), Yamada; Yoshihiro (Tokyo-to, JP), Shin; Tomohiro (Kanazawa, JP)

Applicant: TOYOTA JIDOSHA KABUSHIKI KAISHA (Aichi-ken, JP); MONET Technologies Inc. (Tokyo, JP)

Family ID: 1000008778318

Assignee: TOYOTA JIDOSHA KABUSHIKI KAISHA (Aichi-ken, JP); MONET TECHNOLOGIES INC. (Tokyo, JP)

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Primary Examiner: Manejwala; Ismail A

Attorney, Agent or Firm: HAUPTMAN HAM, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2022-143218 filed on Sep.

8, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to an information processing method.

BACKGROUND

(3) Patent Literature (PTL) 1 discloses that, based on on-demand vehicle reservation information from a plurality of users, an on-demand vehicle operation schedule is determined so that each user can board and alight from an on-demand vehicle at the desired location within the desired boarding and alighting time slot.

CITATION LIST

Patent Literature

(4) PTL 1: WO 2014/002267 A1

SUMMARY

(5) In a conventional configuration, a service area is defined for each on-demand vehicle service, and users of a certain service can only travel by on-demand vehicles within the corresponding service area. Therefore, in a case in which a user wishes to transfer between on-demand vehicles belonging to a plurality of services, for example, the user needs to make separate reservations for on-demand vehicles in each service, which is complicated.

(6) It would be helpful to improve user convenience in the case of transferring between on-demand vehicles belonging to a plurality of services.

(7) An information processing method according to the present disclosure is an information processing method to be executed by an information processing apparatus capable of communicating with a terminal apparatus used by a user, the information processing method including: a first step of receiving, from the terminal apparatus, first information indicating a travel plan of the user; a second step of acquiring, for each on-demand vehicle service in a plurality of on-demand vehicle services, second information indicating a plurality of boarding/alighting points at which a vehicle can be boarded and alighted from and an operation status of the vehicle; a third step of determining, based on the first information and the second information, a recommended route for traveling by transferring between a first vehicle belonging to a first on-demand vehicle service and a second vehicle belonging to a second on-demand vehicle service; a fourth step of presenting the recommended route to the user via the terminal apparatus; and a fifth step of reserving the first vehicle and the second vehicle with the first on-demand vehicle service and the second on-demand vehicle service in accordance with the recommended route in response to a request from the terminal apparatus, wherein in the third step, a first boarding/alighting point for boarding the first vehicle, a candidate transfer point for transferring from the first vehicle to the second vehicle, and a second boarding/alighting point for alighting from the second vehicle are determined, at least one candidate route for traveling from the first boarding/alighting point to the second boarding/alighting point via the candidate transfer point is determined, and based on a result of evaluating each candidate route in the at least one candidate route by a predetermined index, a candidate route included in the at least one candidate route is determined as the recommended route.

(8) According to the present disclosure, user convenience can be improved in the case of transferring between on-demand vehicles belonging to a plurality of services.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the accompanying drawings:

(2) FIG. 1 is a diagram illustrating a configuration example of a system according to an embodiment of the present disclosure;

- (3) FIG. 2 is a diagram illustrating an example of areas where on-demand vehicle services are provided;
- (4) FIG. 3 is a flowchart illustrating an example of operations of a control apparatus;
- (5) FIG. 4 is a diagram illustrating an example of areas where on-demand vehicle services are provided; and
- (6) FIG. 5 is a diagram illustrating an example of areas where on-demand vehicle services are provided.

DETAILED DESCRIPTION

(7) An embodiment of the present disclosure will be described below, with reference to the drawings.

(8) (Outline of Embodiment)

(9) FIG. 1 is a diagram illustrating a configuration example of a system 1 according to an embodiment of the present disclosure. The system 1 includes a control apparatus 10, a terminal apparatus 20, and a vehicle 30. The control apparatus 10, the terminal apparatus 20, and the vehicle 30 are communicably connected to a network 40 including, for example, the Internet, a mobile communication network, and the like.

(10) The control apparatus 10 is an information processing apparatus that receives first information indicating a travel plan of a user from the terminal apparatus 20 and determines an operation schedule of the vehicle 30. The control apparatus 10 may, for example, be a computer, such as a server apparatus, installed in a data center or other facility. The control apparatus 10 can communicate with the terminal apparatuses 20 and the vehicles 30 via the network 40.

(11) The terminal apparatus 20 is an information processing apparatus that transmits the travel plan to the control apparatus 10 in response to an operation by the user. The terminal apparatus 20 is a smartphone used by the user in the present embodiment but may also be a mobile device, such as a mobile phone or tablet, as well as a general purpose device such as a personal computer (PC). The number of terminal apparatuses 20 included in the system 1 may be freely determined.

(12) The vehicle 30 is an automobile, such as a bus, but is not limited to this and may be any vehicle. The automobile is, for example, a gasoline vehicle, a hybrid electric vehicle (HEV), a plug-in hybrid electric vehicle (PHEV), a fuel cell electric vehicle (FCEV), a battery electric vehicle (BEV), or the like, but is not limited to these. Each vehicle 30 is an autonomous vehicle (AV) in the present embodiment, but the vehicle 30 may be driven by a driver, or the driving may be automated at any level. The automation level is, for example, any one of Level 1 to Level 5 according to the level classification defined by the Society of Automotive Engineers (SAE). Each vehicle 30 may be a Mobility as a Service (MaaS) dedicated vehicle. The number of vehicles 30 included in the system 1 may be freely determined.

(13) The vehicle 30 operates as an on-demand vehicle with an operation schedule (operation route, operation time, and the like) determined in response to a user request (demand). The operation route, operation time, and the like of the vehicle 30 are not predetermined, and the control apparatus 10 dynamically determines the operation schedule according to the user's request (demand). The vehicle 30 transmits and receives various information, including information representing the operation schedule, through communication with the control apparatus 10 and travels according to the operation schedule in a specific area defined for each service 111, 112. Hereinafter, the vehicle 30 traveling in a service A111 as the first on-demand vehicle service may be referred to as the first vehicle 30a, and the vehicle 30 traveling in a service B112 as the second on-demand vehicle service may be referred to as the second vehicle 30b. The vehicle 30 is a passenger bus that an unspecified number of users board and alight from but may instead be a vehicle that a specific number of users board and alight from. In the present embodiment, an example is described in which the boarding/alighting points at which the users can board and alight from the vehicle 30 are predetermined, but the boarding/alighting points may also be dynamically determined according to the users' requests.

(14) FIG. 2 is a diagram illustrating an example of areas where on-demand vehicle services are provided. In FIG. 2, the user wishes to travel from a departure point **101** to a destination point **102**. On-demand vehicle services **A111** and **B112** are provided in the area illustrated in FIG. 2. In the example in FIG. 2, the departure point **101** is under the authority of service **A111**, and the destination point **102** is under the authority of service **B112**. Travel reservations for service **A111** and service **B112** must be made separately. In a conventional configuration, users have to make each reservation themselves, which is cumbersome. As described below, in the present embodiment, the user simply inputs a travel plan, and the control apparatus **10** automatically makes travel reservations for each service. Service **A111** and service **B112** are provided by different businesses but may be provided by the same business.

(15) A plurality of boarding/alighting points **121-127** are determined in advance in the areas where the on-demand vehicle services are provided. In the example in FIG. 2, a boarding/alighting point **121** is established in the service area of service **A111**. Boarding/alighting points **123**, **126** are established in the service area of service **B112**. Boarding/alighting points **122**, **124**, and **125** are established in the overlapping portion between the service area of service **A111** and the service area of service **B112**. Facilities **131-134** such as restaurants, public facilities, and schools are located in the area illustrated in FIG. 2.

(16) In the above situation, the user operates the terminal apparatus **20** to make a travel reservation by inputting the first information indicating the travel plan, such as the departure point **101**, destination point **102**, desired reservation date and time, and number of people in the reservation. The terminal apparatus **20** transmits the first information inputted by the user to the control apparatus **10**. Upon receiving the first information indicating the travel plan, the control apparatus **10** identifies a plurality of candidate transfer points where the user can transfer between vehicles **30** belonging to different services and determines a plurality of candidate routes from the departure point **101** to the destination point **102** via each candidate transfer point. Furthermore, the control apparatus **10** presents the user with a recommended route selected based on the result of evaluating each candidate route by a predetermined indicator (such as the number of registered facilities passed by or the travel time in the vehicle **30**) and makes reservations for two on-demand vehicle services according to the recommended route in response to a request from the user. Therefore, even in a case in which a user travels by transferring between on-demand vehicles belonging to a plurality of services, the user does not need to make separate reservations for on-demand vehicles with each service, thereby reducing the burden on the user.

(17) (Configuration of Control Apparatus)

(18) As illustrated in FIG. 1, the control apparatus **10** includes a controller **11**, a memory **12**, and a communication interface **13**.

(19) The controller **11** includes at least one processor, at least one programmable circuit, at least one dedicated circuit, or a combination of these. The controller **11** controls operations of the entire control apparatus **10**.

(20) The memory **12** includes one or more memories. The memories included in the memory **12** may each function as, for example, a main memory, an auxiliary memory, or a cache memory. The memory **12** stores any information used for operations of the control apparatus **10**. For example, the memory **12** may store a system program, an application program, a database, map information, operation information for each service, and the like. The information stored in the memory **12** may be updated with, for example, information acquired from the network **40** via the communication interface **13**.

(21) The communication interface **13** includes at least one interface for communication for connecting to the network **40**. The interface for communication is compliant with, for example, mobile communication standards, wired local area network (LAN) standards, or wireless LAN standards, but is not limited to these, and may be compliant with any communication standards. In the present embodiment, the control apparatus **10** communicates with the terminal apparatuses **20**

and the vehicles **30** via the communication interface **13** and the network **40**.

(22) In the present embodiment, the second information indicating the information on the plurality of boarding/alighting points **121-127** and the operation status of the vehicles **30** in each of the services **A111** and **B112** is stored in the memory **12** of the control apparatus **10**. The operation of the vehicles **30** in each of the services **A111** and **B112** is managed on the control apparatus **10**. Alternatively, the storage of the second information and the operation management of the vehicles **30** may be handled on a network storage or information processing apparatus separate from the control apparatus **10** for each on-demand vehicle service.

(23) (Configuration of Terminal Apparatus)

(24) As illustrated in FIG. **1**, the terminal apparatus **20** includes a controller **21**, a memory **22**, a communication interface **23**, and an input/output interface **24**.

(25) The controller **21** includes at least one processor, at least one programmable circuit, at least one dedicated circuit, or a combination of these. The controller **21** controls operations of the entire terminal apparatus **20**.

(26) The memory **22** includes one or more memories. The memories included in the memory **22** may each function as, for example, a main memory, an auxiliary memory, or a cache memory. The memory **22** stores any information used for operations of the terminal apparatus **20**. For example, the memory **22** may store a system program, an application program, map information, or the like. The information stored in the memory **22** may be updated with, for example, information acquired from the network **40** via the communication interface **23**.

(27) The communication interface **23** includes at least one interface for communication for connecting to the network **40**. The interface for communication is compliant with, for example, a mobile communication standard or a wireless LAN standard but is not limited to these and may be compliant with any communication standard. In the present embodiment, the terminal apparatus **20** communicates with the control apparatus **10** via the communication interface **23** and the network **40**.

(28) The input/output interface **24** is a Human Machine Interface (HMI) that accepts input operations from the user and outputs the processing results of the terminal apparatus **20** to the user. The input/output interface **24** is, for example, configured as a touch screen integrated provided with a display, but this example is not limiting. For example, the input/output interface **24** may accept input operations from the user using physical keys, capacitive keys, a pointing device, a microphone, or the like. The input/output interface **24** may also output information to the user through a speaker or vibrator.

(29) (Configuration of Vehicle)

(30) As illustrated in FIG. **1**, the vehicle **30** includes a controller **31**, a memory **32**, a communication interface **33**, and a positioner **34**.

(31) The controller **31** includes at least one processor, at least one programmable circuit, at least one dedicated circuit, or a combination of these. The controller **31** controls operations of the entire vehicle **30**.

(32) The memory **32** includes one or more memories. The memories included in the memory **32** may each function as, for example, a main memory, an auxiliary memory, or a cache memory. The memory **32** stores any data used for the operations of the vehicle **30**. For example, the memory **32** may store a system program, an application program, map information, or the like. The information stored in the memory **32** may be updated with, for example, information acquired from the network **40** via the communication interface **33**.

(33) The communication interface **33** includes at least one interface for communication for connecting to the network **40**. The communication interface is compliant with mobile communication standards, for example, but not limited to these, and may be compliant with any communication standard. In the present embodiment, the vehicle **30** communicates with the control apparatus **10** via the communication interface **33** and the network **40**.

- (34) The positioner **34** includes one or more apparatuses configured to acquire positional information for the vehicle **30**. Specifically, the positioner **34** includes a receiver corresponding to the Global Positioning System (GPS), for example, but is not limited to this and may include a receiver corresponding to any satellite positioning system.
- (35) Each vehicle **30** runs according to an operation schedule received from the control apparatus **10**. Each vehicle **30** also transmits information on the position of the vehicle **30** measured by the positioner **34** to the control apparatus **10**.

Operation Example 1

- (36) An example of operations of the system **1** is described with reference to FIGS. **2** and **3**. FIG. **3** is a flowchart illustrating an example of operations of the control apparatus **10**. The operations of the control apparatus **10** described with reference to FIG. **3** can correspond to at least a portion of an information processing method according to the present embodiment. The steps in FIG. **3** are performed under the control of the controller **11** in the control apparatus **10**.
- (37) Before the process in the flowchart begins, the user operates the terminal apparatus **20** to input the travel plan. The travel plan of the user includes the departure point **101**, the destination point **102**, the desired reservation date and time, and the number of people in the reservation. Here, the desired reservation date and time may include the desired boarding time and desired alighting time, or may include the desired departure time or desired arrival time. The user may select the boarding/alighting point (for example, boarding/alighting point **121**) at which the user wishes to board as the departure point **101** and select the boarding/alighting point (for example, boarding/alighting point **123**) at which the user wishes to alight as the destination point **102**. The terminal apparatus **20** transmits these inputted pieces of information to the control apparatus **10** as the first information.
- (38) In step **S1**, the controller **11** of the control apparatus **10** receives the first information indicating the travel plan of the user from the terminal apparatus **20** via the communication interface **13**.
- (39) In step **S2**, the controller **11** acquires second information indicating the plurality of boarding/alighting points **121-127** at which the vehicle **30** can be boarded and alighted from and the operation status of the vehicle **30** for each of the on-demand vehicle services **111**, **112**. In the present embodiment, the control apparatus **10** reads and retrieves the second information stored in the memory **12**. The information for the boarding/alighting points **121-127** may include positional information for each of the boarding/alighting points **121-127**. The information on the operation status of the vehicle **30** may include information such as the position and an already reserved operation schedule for each vehicle **30**. In a case in which the second information is stored in network storage or the like for each service, the controller **11** acquires the second information from these network storage locations via the communication interface **13**.
- (40) In steps **S3-S6**, the controller **11** of the control apparatus **10** executes a process to determine a recommended route for traveling by transferring between the first vehicle **30a** belonging to the service **A111** and the second vehicle **30b** belonging to the service **30b** based on the first information and the second information.
- (41) First, in step **S3**, the controller **11** determines, based on the first information and the second information, a boarding/alighting point for boarding the first vehicle **30a** (first boarding/alighting point), a candidate transfer point for transferring from the first vehicle **30a** to the second vehicle **30b**, and a boarding/alighting point for alighting from the second vehicle **30b** (second boarding/alighting point).
- (42) Specifically, the controller **11** may determine the nearest boarding/alighting point **121** from the departure point **101** indicated by the first information as the boarding/alighting point for boarding the first vehicle **30a**. Alternatively, the controller **11** may determine a plurality of boarding/alighting points within a certain distance (for example, X meters) from the departure point **101** as boarding/alighting points for boarding the first vehicle **30a**. The controller **11** may

determine the boarding/alighting points **122**, **124**, **125**, which are established in the overlapping portion between the service area of service **A111** and the service area of service **B112**, as candidate transfer points for transferring from the first vehicle **30a** to the second vehicle **30b**. The controller **11** may determine the nearest boarding/alighting point **123** from the destination point **102** indicated by the first information as the boarding/alighting point for alighting from the second vehicle **30b**. Alternatively, the controller **11** may determine a plurality of boarding/alighting points **123**, **126** within a certain distance (for example, X meters) from the destination point **102** as boarding/alighting points for alighting from the second vehicle **30b**. In the example in FIG. 2, the boarding/alighting point **127** is outside the service area of service **B112** and therefore is not selected as the boarding/alighting point for alighting from the second vehicle **30b**.

(43) In step **S4**, the controller **11** determines at least one candidate route for traveling from the departure point **101** to the destination point **102** via the candidate transfer point. For example, in the area in FIG. 2, the controller **11** may determine the following candidate routes 1-3. Candidate route 1: a route through departure point **101**, boarding/alighting point **121**, boarding/alighting point **122**, boarding/alighting point **123**, and destination point **102**. Candidate route 2: a route through departure point **101**, boarding/alighting point **121**, boarding/alighting point **124**, boarding/alighting point **123**, and destination point **102**. Candidate route 3: a route through departure point **101**, boarding/alighting point **121**, boarding/alighting point **125**, boarding/alighting point **123**, and destination point **102**.

(44) In step **S5**, the controller **11** evaluates each of the candidate routes 1-3 by a predetermined index. As such an index, the controller **11** may, for example, calculate the travel time. For example, the controller **11** may estimate the total time required to travel from the departure point **101** to the destination point **102** and use that time as an evaluation index. The controller **11** may instead evaluate each candidate route using the riding time (time spent in the vehicle **30**) as an index. The controller **11** may instead use the time required to transfer between vehicles **30** at the boarding/alighting points **122**, **124**, **125** as such an evaluation index. The controller **11** may instead use the time traveling by walking as such an evaluation index. These times are determined by taking into account information about the operation schedules already reserved for the first vehicle **30a** and the second vehicle **30b**. Such an evaluation index is not limited to the examples listed here. Other examples of the evaluation index are discussed below.

(45) In step **S6**, the controller **11** determines a candidate route included in the plurality of candidate routes 1-3 as the recommended route based on the result of evaluating each of the candidate routes 1-3 by the predetermined index. For example, the controller **11** may determine the route with the shortest total time, riding time, time required to transfer, or walking time in order to travel from the departure point **101** to the destination point **102** as the recommended route. The controller **11** may instead determine the route with the lowest weighted sum of the total travel time, riding time, transfer time, and walking time as the recommended route. The controller **11** may determine a plurality of routes with a high degree of recommendation as recommended routes based on the evaluation index. Here, the recommended route may include not only a geographic route, but also information on the estimated time of departure and estimated time of arrival at each point.

(46) In step **S7**, the controller **11** presents the recommended route determined in step **S6** to the user via the terminal apparatus **20**. Specifically, the controller **11** transmits information indicating the recommended route determined in step **S6** to the terminal apparatus **20** via the communication interface **13**. In response, the terminal apparatus **20** displays the recommended route on the input/output interface **24** and accepts a selection by the user. When the user selects the desired recommended route, the terminal apparatus **20** transmits information indicating the recommended route selected by the user to the control apparatus **10** and requests that the control apparatus **10** make a reservation for each service **111**, **112**.

(47) In step **S8**, the controller **11** transmits a reservation request, according to the recommended route, for the first vehicle **30a** and second vehicle **30b** to the on-demand vehicle service **A111** and

service **B112** in response to a request from the terminal apparatus **20**. Reservations are thereby made for both service **A111** and service **B112**.

(48) As described above, in the present embodiment, when a user inputs a travel plan, the control apparatus **10** determines a recommended route that conforms to that travel plan and makes reservations for each service according to the recommended route in response to a request by the user. Therefore, even in a case in which the user travels by transferring between the first vehicle **30a** and second vehicle **30b** belonging to a plurality of services **111**, **112**, the user does not need to make separate reservations for the first vehicle **30a** and the second vehicle **30b** with each service **111**, **112**. Therefore, according to the present embodiment, the burden on the user can be reduced.

Operation Example 2

(49) In Operation Example 1, an example in which candidate routes are evaluated based on an index focused on travel time for the user was described, but the evaluation index is not limited to this example. For example, the controller **11** may determine, with greater priority, a route that passes near a facility that is likely to be of interest to the user as the recommended route. By traveling on such a route on a priority basis, the facility can be advertised to the user. This can, for example, increase the likelihood that the user will visit the facility on the way back from the destination point **102** to the departure point **101** or on another occasion.

(50) Specifically, for example, the control apparatus **10** may store information such as the location and category (for example, restaurant, public facility, school, museum, temple/shrine, or the like) of each facility **131-134** in the target area of the on-demand vehicle in the memory **12** in advance. Additionally, in step **S5**, the controller **11** may assign a greater evaluation score to candidate routes that pass more facilities that are likely to be of interest to the user. For example, the control apparatus **10** may store categories of facilities of interest to each user in the memory **12** in advance. The control apparatus **10** may then use the count of facilities that are located near the candidate route and match the user's interests as an evaluation score. The existence of a facility near a candidate route may be determined based on the distance between the facility and the candidate route being less than a certain value.

(51) As a facility that is likely to be of interest to the user, the controller **11** may, for example, select a facility such that the number of times the user has visited the facility in the past or the number of times the user has used a coupon at the facility is equal to or greater than a predetermined threshold or less than a predetermined threshold. Here, the control apparatus **10** may store information regarding the number of times the user has visited the facility in the past or the number of times the user has used a coupon at the facility in the memory **12** in advance. The controller **11** may select a facility that is less than a certain distance from a boarding/alighting point that the user has previously alighted from as a facility that the user has previously visited.

(52) Alternatively, when searching for a route between the departure point **101** or the destination point **102** and the candidate transfer point, the controller **11** may search for the route that maximizes the number of facilities on the route, without the travel time by the vehicle **30** exceeding an allowable value. The “allowable value” may be defined based on the minimum travel time between the departure point **101** or the destination point **102** and the candidate transfer point. For example, the “allowable value” may be the value yielded by adding a certain time α to the minimum travel time between the departure point **101** or the destination point **102** and the candidate transfer point (minimum travel time $+\alpha$). Alternatively, the “allowable value” may be the value yielded by multiplying the minimum travel time between the departure point **101** or the destination point **102** and the candidate transfer point by a certain value β ($1<\beta$) (minimum travel time $\times\beta$).

Operation Example 3

(53) As described above, in step **S4**, the controller **11** determines the routes from the departure point **101** to the candidate transfer points and the routes from the candidate transfer points to the destination point **102** as candidate routes. Here, the controller **11** need not search for a route by

targeting the entire area served by the on-demand vehicle service when determining the candidate routes. For example, the controller **11** may select an area to search for a route in the target areas of the services based on factors such as the positional relationship between the departure point **101** and the destination point **102** and may then search for a route within the selected area.

(54) FIG. 4 is a diagram illustrating an example of areas where on-demand vehicle services are provided. In FIG. 4, the target area of service A**111** includes area **1111** and area **1112**. The target area of service B**112** includes area **1121** and area **1122**. The departure point **101** is located in area **1112** and the destination point **102** is located in area **1121**. It is assumed that areas **1111**, **1112**, **1121**, and **1122** have the positional relationship illustrated in FIG. 4. In such a case, when searching for a route from the departure point **101** to the candidate transfer points **122**, **124**, **125**, the controller **11** need only search for routes in area **1112**, without considering routes in area **1111**. When searching for a route from candidate transfer points **122**, **124**, **125** to the destination point **102**, the controller **11** need only search for routes in area **1121**, without considering routes in area **1122**. In this way, the control apparatus **10** can perform an efficient route search without extracting routes that are clearly circuitous as candidate routes.

Operation Example 4

(55) In step S3 of Operation Example 1, an example was described in which the controller **11** determines the boarding/alighting points **122**, **124**, **125**, which are established in the overlapping portion between the service areas of service A**111** and service B**112**, as candidate transfer points. However, it may be the case that there is no overlap between the service areas of service A**111** and service B**112**. In such a case, the controller **11** may evaluate each candidate route using the travel time by walking from the alighting location of the first vehicle **30a** to the boarding location of the second vehicle **30b** as one index.

(56) FIG. 5 is a diagram illustrating an example of areas where on-demand vehicle services are provided. In the example in FIG. 5, there is no overlap between the service areas of service A**111** and service B**112**. For travel between such service areas where there is no overlap, in step S3, the controller **11** may determine the candidate transfer points based on the positional relationship between the service areas of each service. For example, in the example in FIG. 5, the controller **11** may determine the boarding/alighting points **122**, **125** located near service B**112** as candidate transfer points within service A**111**. Similarly, the controller **11** may determine the boarding/alighting points **124**, **128** located near service A**111** as candidate transfer points within service B**112**.

(57) Alternatively, at least one candidate transfer point may be stored in the memory **12** in advance for each of the travel source and destination services, and in step S3, the controller **11** may determine the candidate transfer points by referring to such information stored in advance. For example, in the example in FIG. 5, the boarding/alighting points **122**, **125** and boarding/alighting points **124**, **128** may be stored in the memory **12** in advance as candidate transfer points between service A**111** and service B**112**, together with identification information for service A**111** and service B**112**. Then, in step S3, the controller **11** may determine the boarding/alighting points **122**, **125** and the boarding/alighting points **124**, **128**, stored in advance in correspondence with service A**111** and service B**112**, as candidate transfer points.

(58) In step S5, the controller **11** may evaluate the candidate routes using the travel time by walking (transfer time) between the boarding/alighting points **122**, **125** and the boarding/alighting points **124**, **128** as one index. Specifically, there are four combinations of these candidate transfer points, i.e., (1) boarding/alighting points **122**, **124**, (2) boarding/alighting points **122**, **128**, (3) boarding/alighting points **125**, **124**, and (4) boarding/alighting points **125**, **128**. The controller **11** may use the shortest travel time by walking among these combinations as one evaluation index. The controller **11** can thereby determine a route that is less burdensome for the user as a recommended route, even in a case in which the route involves travel between non-overlapping service areas.

(59) In step S5, the controller **11** may calculate an evaluation score that comprehensively reflects a plurality of evaluation indices by combining the aforementioned plurality of evaluation indices after weighting them. In other words, the controller **11** may evaluate each candidate route by combining two or more of the above-described indices. For example, the controller **11** may determine the cost for each index and determine the candidate route with the lowest total cost as the recommended candidate. The controller **11** can thereby determine an appropriate route as the recommended route based on multiple perspectives.

(60) While the present disclosure has been described with reference to the drawings and embodiments, it should be noted that various modifications and revisions may be implemented by those skilled in the art based on the present disclosure. Accordingly, such modifications and revisions are included within the scope of the present disclosure. For example, functions or the like included in each component, each step, or the like can be rearranged without logical inconsistency, and a plurality of components, steps, or the like can be combined into one or divided.

(61) For example, an embodiment in which the configuration and operations of the control apparatus **10** in the above embodiment are distributed to multiple computers capable of communicating with each other can be implemented. For example, an embodiment in which some or all of the components of the control apparatus **10** are provided in the terminal apparatus **20** or the vehicle **30** can also be implemented.

(62) For example, an embodiment in which a general purpose computer functions as the control apparatus **10** according to the above embodiment can also be implemented. Specifically, a program in which processes for realizing the functions of the control apparatus **10** according to the above embodiment are written may be stored in a memory of a general purpose computer, and the program may be read and executed by a processor. Accordingly, the present disclosure can also be implemented as a program executable by a processor, or a non-transitory computer readable medium storing the program.

Claims

1. An information processing method to be executed by an information processing apparatus communicating with a terminal apparatus used by a user, the information processing method comprising: receiving, from the terminal apparatus, first information indicating a travel plan of the user; acquiring, for each on-demand vehicle service in a plurality of on-demand vehicle services, second information indicating a plurality of boarding/alighting points at which a vehicle can be boarded and alighted from and an operation status of the vehicle; determining, based on the first information and the second information, a recommended route for traveling by transferring between a first vehicle belonging to a first on-demand vehicle service and a second vehicle belonging to a second on-demand vehicle service; presenting the recommended route to the user via the terminal apparatus; and in response to a request from the terminal apparatus, reserving the first vehicle and the second vehicle with the first on-demand vehicle service and the second on-demand vehicle service in accordance with the recommended route, determining an operation schedule of the first vehicle and the second vehicle in accordance with the recommended route, and controlling, by a communication from the information processing apparatus, the first vehicle and the second vehicle to travel according to the operation schedule, wherein each of the first vehicle and the second vehicle is a fully automated vehicle automated at Level 5 according to a level classification defined by Society of Automotive Engineer, in the determining of the recommended route, a first boarding/alighting point for boarding the first vehicle, a candidate transfer point for transferring from the first vehicle to the second vehicle, and a second boarding/alighting point for alighting from the second vehicle are determined, at least one candidate route for traveling from the first boarding/alighting point to the second boarding/alighting point via the candidate transfer point is determined, and based on a result of evaluating each candidate route in the at least one candidate

route by a predetermined index, a candidate route included in the at least one candidate route is determined as the recommended route, and in the determining of the recommended route, in response to determining (i) no overlap between a first service area, where the first on-demand vehicle service is provided, and a second service area, where the second on-demand vehicle service is provided, and (ii) no service area bridging between the first service area and the second service area, based on a positional relationship between the first service area and the second service area, each of at least one third boarding/alighting point for alighting from the first vehicle in the first service area and at least one fourth boarding/alighting point for boarding the second vehicle in the second service area is determined as the candidate transfer point, and at least one route for traveling from the first boarding/alighting point to the second boarding/alighting point via the at least one third boarding/alighting point and the at least one fourth boarding/alighting point is determined as the recommended route.

2. The information processing method according to claim 1, wherein in the determining of the recommended route, the candidate route included in the at least one candidate route is determined as the recommended route based on a result of evaluating a total travel time as the predetermined index.

3. The information processing method according to claim 1, wherein in the determining of the recommended route, the candidate route included in the at least one candidate route is determined as the recommended route based on a result of evaluating a count of facilities that are located near the candidate route and match interests of the user as the predetermined index.

4. The information processing method according to claim 1, wherein in the determining of the recommended route, the candidate route included in the at least one candidate route is determined as the recommended route based on a result of evaluating a transfer time by walking as the predetermined index.

5. The information processing method according to claim 3, wherein the facilities include a first facility to which the user has visited in the past, and a second facility at which the user has used a coupon in the past, and each of the first facility and the second facility is weighted greater for evaluating the candidate route than other facilities among the facilities.

6. An information processing method to be executed by an information processing apparatus communicating with a terminal apparatus used by a user, the information processing method comprising: receiving, from the terminal apparatus, first information indicating a travel plan of the user; acquiring, for each on-demand vehicle service in a plurality of on-demand vehicle services, second information indicating a plurality of boarding/alighting points at which a vehicle can be boarded and alighted from and an operation status of the vehicle; determining, based on the first information and the second information, a recommended route for traveling by transferring between a first vehicle belonging to a first on-demand vehicle service and a second vehicle belonging to a second on-demand vehicle service; presenting the recommended route to the user via the terminal apparatus; and in response to a request from the terminal apparatus, reserving the first vehicle and the second vehicle with the first on-demand vehicle service and the second on-demand vehicle service in accordance with the recommended route, determining an operation schedule of the first vehicle and the second vehicle in accordance with the recommended route, and controlling, by a communication from the information processing apparatus, the first vehicle and the second vehicle to travel according to the operation schedule, wherein each of the first vehicle and the second vehicle is a fully automated vehicle automated at Level 5 according to a level classification defined by Society of Automotive Engineer, in the determining of the recommended route, a first boarding/alighting point for boarding the first vehicle, a candidate transfer point for transferring from the first vehicle to the second vehicle, and a second boarding/alighting point for alighting from the second vehicle are determined, at least one candidate route for traveling from the first boarding/alighting point to the second boarding/alighting point via the candidate transfer point is determined, and based on a result of evaluating each candidate route in the at least one candidate

route by a predetermined index, a candidate route included in the at least one candidate route is determined as the recommended route, in the determining of the recommended route, the candidate route included in the at least one candidate route is determined as the recommended route based on a result of evaluating a count of facilities that are located near the candidate route and match interests of the user as the predetermined index, the facilities include a first facility to which the user has visited in the past, and a second facility at which the user has used a coupon in the past, and each of the first facility and the second facility is weighted greater for evaluating the candidate route than other facilities among the facilities.
